

ActInf Textbook Group ~ Cohort 1 ~ Meeting 18 (Chapter 7, part 2)

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hello cohort one

it's meeting 18 we're in our second

discussion on

chapter seven

let's go to the questions and see what

we can explore today

or see where else to go

and or

look towards chapter eight but there's

any number of ways we can do it so

first just on chapter seven

does anyone want to uh

turn to any of these questions add

another question add another

reflection or a thought that arose in

the last week

one question that I um

had was

are there any ways or sweets

for for example taking in an analytical

expression

and then providing equivalent

phrasings

that might have other advantages

Ali

you mean by translating it into natural

language

or like

like what we saw in equation 2.5 and 2.6

like to take in an expression

and then

output

[Music]

um

isomorphic or I guess not isomorphic but

equivalent expressions Expressions that
have the exact same
value as calculated
because some some of the and when we
look through the derivation sometimes
it's possible to trace the trail but to
know which representations of perhaps
even the same
functional
or same
term
it seems quite relevant
actually there's a an AI assistant for
deriving formal proofs for mathematical
theorems but uh I haven't used it myself
but I'm not sure if that's what you mean
I'll look at the
book the name of that and what just one
second I'll post hmm
maybe there's a way to
um
reformulate
these
um specific terms from like a Bayesian
mechanics
uh
formalism like posterior predictive
predicted entropy
that looks like something that Dalton
probably described as well
the question is whether you can just
interchange it
yeah does working with the particular
partition
enable equations to be operated with
more composably
because we can know that
there are certain

operations that are
um like always sometimes never going to
be valid
I'm just kind of asking I don't even
know if that's the threat with a habit
um
so last time we talked primarily about
the
um
Mouse
in the maze
and we're going through
uh the way that the this the chapter is
layering on
features of the model
so first we saw
the mouse just go for it
and now we're going to be uh it gets
where it gets the queue
and this comes to our earlier points
about the resolution of the explore
exploit
trade-off
foreign
anyone can give a thought while I'm
adding it what are posterior predictive
entropy and expected ambiguity
the part of the decomposition of
epistemic value
foreign
let me just make sure to say the
posterior
predictive entropy
is the expected surprise or the entropy
of the distribution of observations
conditioned on a policy
so we're in expected free energy world
we're talking about

um
evaluating policies
with respect to now putting aside
pragmatic value we're talking about
decompositions of how the informational
or the epistemic
value of a policy is evaluated
so there's two terms here
first
how dispersed
is your distribution of outcomes
for that policy
one can imagine that all things being
equal you would want to select policies
that have a tighter distribution of
outcomes
here
we have an expectation interestingly
um this is *italics e*
but it's not a fancy e
do people think there's a difference
that matters or do they think that's a
slight error
I think there should be a typo yeah I'm
going to add it to erata
e is
italics
should be fancy e
okay
it's the expectation
over our hidden State estimates
condition on a policy
and it's the expectation of
the expected ambiguity so
entropy
of
the a matrix
functionally how outcomes depend on

States

so this is saying I want to be

I want to be more

um

is it the case that it says I want to be

more certain

about observations

and how they uh map to policy

and

I want to have a tighter a matrix

Eric

so um knife question why is there an

expectation on the right and now on the

left

I was also going to ask this isn't

entropy already the expectation of

surprise

foreign

just just so another another point of um

clarification for myself that

I want to make sure this is right is um

the um the O tilde is observations

and what we're looking at here is a

functional

Q

so that's something that's

that's a distribution

of

observations that will get over policy

and that's what's being adjusted in the

free energy so we're going to explore

the uh the distributions Q

and the $\tilde{O}$'s are going to be

um

past observation so we're not not

looking forward there we're looking

past because those are actual

observations they're not predicted

through

um the the hidden States s

so straighten that out

are we sure that oh our only past

observations I believe that past

observations are accounted for

with respect to how they influence

the prior

um

but does otilda also include

future observations

well if so then it has to be an

expectation over something that mustn't

it

like but this is conditioned upon a

given policy

then

these are the this is the time series of

observations

expected under that policy

is the left term

the Q expected

distribution of

observations conditioned on policy Jacob

I was just going to uh say that I think

I've seen uh this in other parts of the

book as well and I think they use the

H and then just the expectation notation

for entropy kind of interchangeably

um okay if they do but then isn't this

like a double expectation

this is the this is the expectation of

surprise on the queue but this is the

expectation

of the expectation of surprise

yeah uh

that I don't know

Eric

okay okay but the left term

just um yeah Ali please

uh well shouldn't that first line uh I

mean the H the both

H is

shouldn't both of them

be as the argument of the expectation

according to equation 7.4

because in equation 7.4 we have the

negative epistemic value

but it's somehow different from this

equation yeah that's it

okay

so in 7.4 we see a re

statement

of

expected free energy equation two six

let's leave pragmatic value to the side

for now

here

we see fancy $e q s \tilde{\pi}$ so this is

looking like this probably could or

should

we also see $h p o s h p o s$

minus

$HQ o \pi$

okay so let's take the negative of the

first line

the negative of the first line

you can kind of just flip the terms

it's the expected ambiguity now minus

the the posterior predictive entropy

then

an expectation is being taken over that

oh sorry sorry my bad I felt the the

square brackets I mean it was

encompassing the whole expectation term

sorry sorry

no no you're you're right actually um oh
well there's a lot of layers of it
I I always write as it's written okay
okay yeah yeah
um

okay wait

I always color my brackets when I'm
trying to do this um okay the the
outermost bracket there actually may be
a bracket I don't even know let's try to
figure it out the outer bracket starts
here on the left and it and then it
closes okay pragmatic value if we take
that as a standalone
if we take that as standalone
is anyone else seeing a bracket sort of
scenario

if we look at the brackets as a whole
this first bracket closes this right
Mouse uh I don't think so the first
bracket closes at the end of the S tilde
yeah okay okay yeah parenthesis
okay okay bracket yeah that's okay well
I was confused too I got I I agree that
was I was thinking should we evaluate
the brackets locally
but yes they they are
okay good

okay so it's exactly the negative of
equation seven point eight left side
here is the negative of

so

the first term

um let's just say it in terms of picking
good policies

we want

policies are better with more clarity
about observation outcomes that's the

first term

and a tighter a matrix

that's the second term

okay

middle line

this is a KL Divergence

between

the product two two products of

distributions

both of them are dealing with

o

in the first and s in the second

the Q distribution in the second part is

the same for both

so maybe there's some interactions in

how distributions of different kinds

multiply

but

it may be fair to interpret this as um

the POS

is the a matrix

and the $q_{\theta}(\pi)$

is how observations are

shaped by policy

or another way to say it might be

POS

is outcome expectations conditioned upon

hidden states of the world

whereas this is outcome distributions

conditioned on what we do

I mean to me that just looks like like

base you know base law there because

you've got

um at least in the

in the left side a little bit because

you've got condition on S and then s

conditioned on π

so you're just

basically saying you're trying to
figure out what θ is from PI through
this intermediary s Delta
oh one so
another angle would be where's this zero
well this is zero
when
these two are identical
when
the epistemic value of a policy is low
when it when the
resulting θ distribution
is being unconditioned by the policy
that we choose
whereas for policies that change how we
think about the a matrix
those policies have higher informational
value
because POS is kind of like a policy
agnostic mapping
but we're like drilling down to kind of
sub variance partitions
of informational value these are
different ways to look at
what makes a policy
have epistemic value
the first one line was coming down to uh
clarity
around the tightness of outcome
distributions
the second one is related to
how
policy
deviates our understanding of the a
matrix
okay
now let's turn to
okay here's another

um italic e not a fancy e

thank you Ali for this link I've I've

heard of this one the the proof

assistant but I've never seen it used or
anything

yeah

this looks pretty pretty interesting I

wonder if anyone has done that kind of

formal work on proving certain parts of
fep

and then a more distal

um

question

would be if we have formal ontology

representations for example with sumo or

otherwise

could we make proofs

around the ontology

and the Sumo

um suggested upper merged ontology

we explored some of those things but

didn't didn't go super super far

it was just the the it's it's it yeah

take somebody who knows that area

okay

the bottom term the bottom line

an expectation over q but now our Q

is on outcomes

not

focusing on the hidden States

then it's an expectation

about outcomes

of

a kale Divergence

between

it's about this is about hidden States

this kale Divergence

is zero

when outcomes

are not

when in addition to policy outcomes

don't change how we think about hidden

States

so when outcomes influence how we think

about hidden States

this is going to be higher

and what

what does on the right

this is a triangle meaning defined as

so $q(s|o)$

is defined as

and this looks like a Bayes theorem

situation

like kind of that chaining that Eric

just mentioned

like you have $P(O|S)$ condition on S and

then you have S conditioned on π

so then those kind of like

it's like a way to get from o through S

to π

whereas this one on the bottom is just

like going directly about o conditioned

on π

which interestingly is I don't know if

this is reflected one of these or or in

some subsets of them

um one can think about that kind of

structural model like the minimum hidden

Markov model not even in its full

framing but just like what are the three

pillars of the Hidden Markov model

observations hidden States policy

yes Transitions and so on and so what is

gained by going through an intermediary

hidden state

there's going to be situations where the

observable contains a hundred percent of
the information for policy
or we can imagine situations where the
ability to choose policies
based upon s
which is where the actual pomdp is
intervening
is better
whereas in a fully observable Markov
decision process
you don't have S
you're just making decisions on o
um kind of just revisiting the point I
raised earlier about what o tilde is
it seems to me that this only works
when you can when you have the
observations o h tilde
so
I I don't see how that could how it
could be anything like looking toward
the future
because if you actually want to carry
out that um
that expect that that you want to work
out what Q is
or use it you have to compute it and you
have to have the observations to compute
it and and you know the S tilde you have
to have
you know actually pipe through
and
use them so help me help me understand
if I'm wrong with that
okay here would be
um one thought the A matrix captures how
hidden States map to observations
so whether we have a really tight A
matrix or whether we have a super

dispersed a matrix
using the gaussian form for the a matrix
ensures that we can always say what the
most likely observation is
for a given hidden state
if we had a bimodal a matrix
then
there are some issues
but if we use a gaussian form
for the um like which is the laplacian
approximation
it's more tractable it's monotonic it's
more optimizable
so for a given s sequences of s
we can always derive
the expected observations
again those might have a lot of
uncertainty around them but we can
always make a sequence of O's
corresponding to any sequence of s
but this this equation makes no
expectation at all nothing says nothing
about what the form of the distribution
is that you use I mean whether it's
parametric or not parametric or whatever
it's just saying you have a distribution
Q
um
so you know then this is how we evaluate
it
yes is this true for all families of
distributions
Q
Ali what do you think about that or
anything
uh actually it's interesting that uh in
the step-by-step active inference paper
this epistemic value is defined uh

somewhat differently uh I'm not sure if they're exactly equivalent with each other or there's some minor uh differences between them but uh they've just

defined the epistemic value only in terms of the expectation over I mean the surprises of I mean the difference between the q_{np} surprises so we don't have

um
this kind of expression as conditioned on the S tilde here so
yeah that's

I don't know if they're the same or not
I don't know how to copy the image into this chat but the chat of The Gather but or just put it put it in here put it in the um

question

okay yeah

um well one

one thought is policy is not an open-ended policy in its specification is always going to be of a finite Horizon so if we're considering policies of length 2.

um well so so interestingly this is a policy that includes leading up to the present

if this is the present

I think we had one meme that was like what if my T minus one was was t or something like that but

um this is a policy that right is influencing

and this is just the variant of four

three

okay but let's just say that we're
considering policy of true future
looking length 2.

then

we're exploring different ways in which
s is going to unfold
with respect to basically B1 and B2
not using that in the superscript way to
mean like two two parallel bees but just
like there's our current there's our
estimate now

and then there's the way that that we
have you know let's just say four
affordances so there's four B ones
and then

you have now there's going to be four
options for S and then each of those
four we could apply the second
affordance again

so there's 16 policies to evaluate
each one is defined by either taking
like B1 B3 B1 B1 B4 B4

and for each of those s
in the next two time steps

we can emit
observations

Ollie

and then Rohan

sorry reminder my hand just stay away

I'm sorry

rod

my

hello yes greetings

yeah so uh going back to what the
posterior distribution is right so how
far

so is it just that isn't it just that we

assume there's a posterior distribution
of states

like an ideal posterior distribution

that we want to

that free energy that the free energy

principle moves moves the current

empirical distribution towards isn't

that I thought that's what we were going

from

the context in chapter one yes so just

just a question yeah go ahead yeah

uh yeah sorry so sorry the cohort one I

think that's what we discussed right so

it's essentially uh trying to maintain

some use races so whatever policies

uh who have performed should eventually

bring the system to that ideal

homeostatic perspective right so uh why

couldn't we just assume like a normal

distribution so even if it is fat tailed

for example if the empirical

distribution is like the key uh

distribution in this

uh formula is you know has a heavy left

tail you can essentially bring it back

towards that by essentially modifying

the policies I thought that that was the

context of this right

yes broadly you're right so when we look

at the full free energy expected free

energy formalization so here's equation

7.4 or equation 2.6 you're absolutely

right that there's a pragmatic value

that's based around reducing Divergence

between observations and preferences but

in equation the one we've been focusing

on and trying to really unpack because

again also like otilda is used in

pragmatic value

we're only talking here I mean they

would still be

yeah yeah but but that would make sense

right because uh your your conditioning

on your actions having some

uh effect on the state on your own State

and your state in the world or maybe

some state in the world so you have to

continuously monitor until it comes down

to the expected observation that's what

the generative model would spit out

right so it makes perfect sense and

it's completely coherent in that way

right

yes I

hope and slash expect and prefer that to

be true what you're describing about

like bringing the homeostatic variable

closer to preferences expectations

that is pragmatic yeah but that would

also yeah so that wouldn't it wouldn't

be surprising that hotel does that and

not be distribution as well as what I'm

saying no you wouldn't need it to be

there yeah you're right it is needed to

be there because these are surprises and

entropies and so on about observations

um and then these this part is really

digging into

um decompositions of epistemic value

so these are

there might be two policies that are

both expected to bring your temperature

down

or whatever

but they have different epistemic values

does it matter if it just brings your

temperature down well it would look the same

um essentially it would have the same effect so why does it matter

uh why why uh did we have to have this kind of uh yeah so there will be

multiple policies that would bring it down bring the temperature down if

you're if the preferences to bring the temperature down it does not matter

which policy is chosen right so that's essentially what I think it's being said

okay so yeah as long as you minimize the expected here's two investment

opportunities from a hundred dollars 150 plus or minus one or 150 plus or

minus 200. which one do you want

that's not what you were discussing

um so it is because they have space is

very different no it's not the same

thing they are the payoff space expected

ambiguity in the 200 plus or minus 200

is very very different or and your

Mutual information will be very

different because the payoff space is

different exactly you are essentially

taking yeah but uh if something is going

to bring down uh temperature plus or

minus two plus or minus four I mean it's

not a huge difference as long as it

brings it down right so that that's

essentially what I'm getting at so okay

why does it matter okay so in this in

this modeling the body's expected

temperature is like the expected return

on investment

and the variance around the expected

temperature is like the variance around

the S uh Investments

and so right but they wouldn't be the same policy is what I'm saying sorry they won't be the same payoff space in the second case so if something uh generates a hundred dollars but has uh a low probability of going bus so that there's a high variance that's not the same payoff as something that reduces the same hundred dollars but is that has a lower variance the payoff space spaces are different and that will be captured in that uh coolback Divergence that you have there

that's right I would prefer the one with the lower that's right that's right because the payoffs yeah so uh but uh I think the discussion was around if multiple things lead to the same epistemic value and they have the same payoff why does it matter which one is taken rate is that last piece there might be multiple policies with the same pragmatic value

and then then epistemic value is the Difference Maker that favors policies that have more clarity about observation outcomes

right right

so pragmatic value would say would would say that's not necessarily true right uh that's not necessarily true let's say that uh okay when the payoff space is completely unknown that the steps that you're taking uh at T equals one two three will bring down the uh we'll we'll have the same pragmatic value but one of these actions will lead to some very bad

consequences that step peoples 15
because you're giving up something else
because if we're talking about
temperature we're in the space of
temperature okay so yeah
all right all right let's uh so let's
just bring it back to the finance okay
the space is dollars that's the dollar
space
uh yeah okay so we could use dollars
numeric fine so we benchmarked the
dollars how many dollars in making
profit so I have two Investments uh one
uh both of them generate a hundred
dollars but one of them requires me to
borrow a hundred dollars in order to
generate a hundred dollars the other one
will generate a hundred dollars over say
50 time steps this would be quicker
the second one where you borrow 100 so
it would be quicker because you're
borrowing a hundred dollars but there's
an interest rate that you have to uh pay
out so there's a much uh so you will
have to generate much more than a
hundred dollars in order to for this to
be viable okay because you have an
interest rate payout okay I'm gonna try
to follow up here policies are defined
in terms of sequences of affordances to
take
so the affordance here that you're
highlighting is borrowing or not
yes but there are actions that would be
like if you have a very high fever in
order to bring down that high fever
sometimes it makes sense to borrow that
hundred dollars whatever metaphorical

variation of that borrowing that you have to bring it down to something more manageable right

so it depends on how close you are to some sort of threshold where it's intolerable

uh I mean putting it in in more mundane terms that hey I have I have a bad toothache it doesn't make sense for me to borrow a hundred dollars to fix my tooth I could just brush my teeth and hope that it goes away but I have a knife in my back I need to get to the hospital

it makes sense to pay a hundred dollars because you're gonna die if you don't look okay let me try to try to see this is how I'm seeing that is when we're um near homeostasis

then yeah we take actions that basically keep us there and we try not to like add risk to the situation whereas near the limits of our homeostatic tolerance we may engage in high risk behavior because there's like already a non-negligible like because of payoff space is much larger yeah the payoff space the positive payoff based on life so when you are when you're closer to when you're closer when you have a knife in your back

the payoff from borrowing a hundred dollars will save your life

but you have a toothache borrowing a hundred dollars might cause more risk that you go bankrupt right but you won't you will still survive if you have the tooth a toothache you're just short a

hundred dollars which you could have
used elsewhere

there's no way you can use that hundred
dollars if you're dead yeah so it does
it does a much higher payoff borrowing
at that point than it is with the
toothache example rates it depends on
where you take these actions I got you
I'm seeing the space

are the axes so here we have dollars and
life

and so you're saying that yeah but it is
a multi-dimensional uh this one so if
you think about it as let's say yeah so
uh let's make it more concrete right so
if you have a drone

uh and uh you have a trade-off between
battery life and staying in the air

uh and let's say one of your propellers
goes out uh you know just redistributing
power to the other motors so that you
stay in the air even though it reduces
your uh hover time makes more sense
because you could Land versus you know
uh trying to fly around and then crash
eventually

because you've lost one motor

so but whereas let's say that uh it's so
the other option would be let's just go
faster to a certain destination

then it doesn't really make sense to do
so you're just reducing your battery
life to get to a destination faster but
uh

but if something happens on the way
which causes you to lose your motor then
you don't have enough battery in order
to enough battery power to distribute to

the automotive so that you can hover and
come down safely

uh does it make sense

uh Eric so this

you know I wonder if we could table this
particular discussion it seems to be
going

um you know and get back to some of the
other questions in the chapter I mean if
I I actually we we had a question that
was pending from last week
that I think is kind of critical to the
understanding of the chapter and I
wonder if we could so you want to make
sure we have time for it running out of
time here yeah

um

and that's the the question about
um

what the relative role is of alert how
it actually operates that you learn
something that the rat learn some
information

prior to deciding which of the two
branches of the maze to take

um and the way they seem to frame the
chapter is that well they set it up so
that the rat

um has its preference
for epistemic Value because of epistemic
value of learning which of the two
branches to take

um is a is a bigger term it'll first go
to the bottom branch

and

um learn which of the two top branches
has the payoff then once that's resolved
it knows that then it go then it decides

which of the top branches to go to and
it gets its reward
so my understanding of that is that
and this is my claim in this question
is that
um the rat is
has a preference to resolve epistemic
uncertainty
regardless of whether it's useful or not
so if you added more uncertain questions
like you know um
what color is
um you know my Apple today but there's
no there's no apple to be gotten
um
it'll still want to resolve that
question and any other questions because
those are uncertainties so the epistemic
value it'll collect that epistemic value
and then finally it gets around to
collecting its reward
um that's in contrast to a model where
the purpose of resolving epistemic
uncertainty is to gain the reward which
requires look ahead so in other words
the the planner or the policy would be I
try to learn what's needed to collect
the reward I do that and that's posted
as as epistemic value with a purpose
then the the order of operation is okay
I figure out what I need to know I learn
what I need to know then I go and get
collect do the exploitation so I don't
that's what I would expect that this
kind of a framing would give us but I
don't see that that's what you're
actually giving us
all right

here here's how I see that the um
example that we would want to see
would have
a bunch of
um uncertainty resolving this one picks
a number between 1 and 10 and tells you
what it is this one's a number between
one and a million so it's an it's an
incredible information resolving button
then the question is can the relevant
sources of information be sought after
um

I think uh the examples is prepared in
several ways
for example the rat already knows the
semantics
that this is related to here
that's been just the structure of the
model

is already preparing that any any
information
any uncertainty reduction
about this like basically it's already
encoded in the model that this bottom
cue is information about this
so this

is kind of like saying
um and this relates earlier like
not only are the those those like
implicit structurally encoded
aspects of knowledge
but also there's parameterization
questions like we discussed
if it really pers if it has a dire
urgency
for food

to um the earlier questions about like
the urgency of the imperative it may

just go for it
and say I'd rather have a 50 50 shot now
than a 98 in two time steps
so it depends on how it's parameterized
but even more deeply it depends on the
construction of the model
what kinds of relationships are
implicitly and explicitly being linked
and like that's a a it's like a deep
level of
modeling
to understand what would be like a more
neutral way
to frame
this question and it's like well how
many layers back do you want to pull
like you could have a pre-play
where a rat has um an association Matrix
there's three information sources
and it's allowed to freely explore
without any shock or food and determine
which one of these information sources
has like a causal relationship to these
edges and then there's a learned cue
that now we're in game time
and now it's going to rely on its past
learning about which information
resource is relevant for the pragmatic
value
and then it's I mean there's just how
many layers back does one need to pull
before the rabbit is not in the Hat
I think is going to be a serious
question
because even toy demonstrations have
been seen as slam dunk
and
um they're not

well instead of bringing up Palm DPS
here Palm DPS are really good for
planting but they don't have this rat do
any planning it seems like they're
really not
aspiring very strongly to build a smart
rat here and they're not achieving it
either
yeah it's um true because
uh
here
the first movement again
presuming that the parameterization is
such that epistemic value is salient
the first move is dominated by epistemic
value and they say well now that that
value has been tapped now it can step
into pursuing pragmatic value with even
increased confidence
um
but it's not considering
this the set of time Horizon 2 policies
that is not being explicitly encoded
or
that means that even implicit yeah these
are these are transition matrices
these are just the four options this is
this is depending on where you are
you can if you're you know you can stay
you can go down
yeah does section seven three address
planning
and if it doesn't why do we need G
why can't we just use a one-step
variational free energy approximation
what's it's it's saying what's the most
likely thing for me for me to do right
now

is a question about now in the past
well I think the answer to the question
is exactly as you said it before which
is they baked in this particular
hardwired rat for which in this case it
it the best thing to do is to First
explore learn and then exploit so they
designed it to do only this very by
wrote to do the right thing as opposed
to the rat actually
use it doing any look ahead or any
intelligence to do the right thing yeah
it's structurally implicitly hardwired
and then it's fine-tuned
because again there's structures where
it wouldn't do this and there's tunings
where it wouldn't do this
but there's there's um
it's cut kind of in in the tens place
and in the decimal point
this example exists in a very limited
manifold
of models and parameterizations where
one step optimal policies emulate
two-step planning
like I just happened to love sacrificing
pawns and taking castles so I'm willing
to sacrifice upon so that later I can do
a castle but of course you can't walk
around with the belief that you love
sacrificing Pawns
yeah
I actually wondered why there was an
enumeration of policies in the planning
section
um
okay
learning hyper priors

hidden States on hidden States
seeing hidden States as outcomes
of other
hidden States
so that they can be learned or fixed
a little bit of technical details on
um on a multi-parameter minimization
Theta being just a vector of parameters
for the generative model
and dear Schley distributions which I
think
could be gotten into but that's kind of
distribution specific
creatures select the most appropriate
data to improve their generative models
I think there's more that we can dig
into and explore on similar wavelengths
like
risk plus ambiguity but then now there's
an Information Gain
I thought we just had risk plus
ambiguity
then we didn't really
yeah people point to structure learning
but don't they always
as a way to get around these questions
but then it's the structure of the
structure learner
and then people end up with these Turing
machines
that don't plan
people hope that Bayesian model
reduction
will be attractable and provide
heuristics for structure learning
but I haven't seen any empirical
examples that come to mind
where Beijing model reduction was used

to identify actionable
lower dimensional
useful models
but we know that structure learning on
the state space of hierarchical models
is going to be essential active
inference or Beyond because
people thought that the explosions the
the computational complexity class of
just branching time active inference or
anything within a model
this is going to be like exploding upon
that by several exponents
and we have these fundamental questions
about the continuous time interpolations
why are there continuous interpolations
in chapter seven
but it's the discrete Time chapter
okay
next time we come to chapter eight I
think this will be quite interesting
we're going to talk about dynamical
systems motor control
um lockabletera we'll have some
Justified continuous lines
um LaPlace assumption Lorenz live stream
number 32 stochastic chaos and Markov
blankets
Hybrid models with discrete continuous
Fusion
and
some advances in continuous time
modeling
thanks everybody
see you soon
thanks everyone